

CHENYE YANG

📍 545 Bainer Hall Dr, Davis, CA 95616 ✉ cyyyang@ucdavis.edu ☎ (917) 215-3998

🌐 github.com/YangChenye 🔗 linkedin.com/in/chenye-yang 🌐 yangchenye.com

EDUCATION

University of California, Davis - United States

Ph.D. Candidate in Electrical and Computer Engineering

Sep 2022 - present

Degree GPA: 3.87/4.0

Columbia University - United States

M.S. Advanced Research Specialization in Electrical Engineering

Sep 2019 - Apr 2021

GPA: 3.8/4.0

Nikola Tesla Electrical Engineering Scholarship

Xi'an Jiaotong University - China

B.Eng. in Automation

Sep 2015 - Jun 2019

Grade: 89.42/100

Honors Electronic and Information Engineering Program (QianXuesen Class)

RESEARCH EXPERIENCES

🌐 full list on website

Robustness of DPO

Prof. Lifeng Lai, UC Davis, May 2025 – present

- Research on adversarial poisoning robustness of Direct Preference Optimization (DPO).

Preference Robustness of RLHF

Prof. Lifeng Lai, UC Davis, Dec 2023 – May 2025

- Research on preference robustness of RLHF algorithms with online human feedback, from mathematical aspects.
- Investigate human feedback attacks on online RLHF: attacker forces the victim algorithm to eventually learn a suboptimal target policy with small attack cost.
- Propose an adversarial human feedback poisoning strategy, and prove its success in misleading the online RLHF.
- Propose a robust defense online RLHF algorithm, and prove it is robust to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. (*Published on IEEE TSP 2025*)

Reward Robustness of Bandit

Prof. Lifeng Lai, UC Davis, Sep 2022 – Feb 2024

- Research on reward attacks on stochastic multi-armed bandit with non-stationary environment, mathematically.
- Investigate attackers who force the victim algorithm to choose a suboptimal arm mostly at a small attack cost.
- Consider three general scenarios with different environment, algorithm, and information available to attacker.
- Propose three attack strategies, and prove they are successful in terms of target arm selection and attack cost.
- Propose a defense non-stationary bandit algorithm, and prove its robustness to any attacker with bounded attack cost.
- Simulation results validate the theoretical analysis. (*Published on ACSSC 2023, IEEE TSP 2024*)

Fever Screening System

Prof. Xiaofan (Fred) Jiang, Columbia University, Jan 2020 – May 2021

- A low-cost RGB–thermal camera system for continuous, contactless fever screening of multiple people; deployed in a NYC restaurant and hospital clinic.
- Investigate and implement key vision modules: RGB–thermal head matching, YOLOv3 for head detection, FSA-Net for head orientation regression, and distance estimation with non-identical RGB/thermal cameras; train, deploy, and evaluate models in real settings. Support end-to-end system validation and deployment.
- Publications accepted by *SenSys 2020 Poster*, *CPHS 2021*, and *IPSN 2022*. Links: Columbia news, project website.

WORK EXPERIENCE

WiLO Networks Inc. Network Engineer Intern

Oct 2021 – Apr 2022

Early-stage startup; owned end-to-end development of the smart-farm prototype across hardware, network, and cloud.

- Architected and implemented a low-cost LoRa mesh network on ESP32 + SX1276 (LoRa PHY), including auto-join/routing protocol and robustness strategies; validated via Duke Smart Farm field tests and NS-3 simulation.
- Integrated and modified MCCI LoRaWAN LMIC at the register level to control SX1276 mode transitions, interrupts, and Tx/Rx scheduling; aligned with ESP32 OS. Job timing for reliable communications.
- Delivered sensor-to-cloud data pipeline (sensors → MCU → LoRa → Raspberry Pi gateway → InfluxDB/Azure) enabling data fusion and ML-based monitoring.
- Standardized reusable MCU libraries + KiCad schematic/footprint libraries for team-wide reuse, and designed and assembled a backscatter-tag PCB in KiCad (component selection + layout).

PUBLICATIONS

- **Chenye Yang**, Mo Lyu, Guanlin Liu, Lifeng Lai, "Human Feedback Attack on Online RLHF: Attack and Robust Defense" *IEEE Transactions on Signal Processing* 2025.
- **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Stochastic Bandits With Non-Stationary Rewards: Reward Attack and Defense" *IEEE Transactions on Signal Processing* 2024.
- **Chenye Yang**, Guanlin Liu, Lifeng Lai, "Reward Attack on Stochastic Bandits with Non-stationary Rewards" *AC-SSC* 2023.
- Kaiyuan Hou, Yanchen Liu, Peter Wei, **Chenye Yang**, Hengjiu Kang, Stephen Xia, Teresa Spada, Andrew Rundle, Xiaofan Jiang, "A Low-Cost In-situ System for Continuous Multi-Person Fever Screening" *IPSN* 2022.
- Stephen Xia, Rishikanth Chandrasekaran, Yanchen Liu, **Chenye Yang**, Tajana Simunic Rosing, Xiaofan Jiang, "Demo Abstract: A Drone-based System for Intelligent and Autonomous Homes" *SenSys 2021 Best Demo Award*.
- Peter Wei, Yanchen Liu, Hengjiu Kang, **Chenye Yang**, Xiaofan Jiang, "A Low-Cost and Scalable Personalized Thermal Comfort Estimation System in Indoor Environments" *CPHS* 2021.
- Peter Wei, **Chenye Yang**, Xiaofan Jiang, "Poster Abstract: Low-Cost Multi-Person Continuous Skin Temperature Sensing System for Fever Detection" *SenSys 2020 Poster*.
- Pengyuan Liu, **Chenye Yang**, Jiang Wu, "Hybrid Features Based K-means Clustering Algorithm for Use in Electricity Customer Load Pattern Analysis" *Chinese Control Conference* 2018.

PROJECTS EXPERIENCES

 full list on website

- Error Correcting Code** *Prof. Ulrich Jetzek (from Fachhochschule Kiel), UC Davis, Apr 2023 - Jun 2023*
- Read TXT, PNG and WAV to bit-stream. Encode with (7,4) Hamming Code and (n, k) Systematic Cyclic Code.
 - Pass Binary Symmetric Channel. Decode with Syndrome and Error Trapping Decoder. Reconstruct file then analyze.
- Sparse Models for High-Dimensional Data** *Prof. John Wright, Columbia University, Jan 2020 - May 2020*
- A system able to fast detect moving objects in high-resolution video. Tested on a sport video and daily scenario.
 - Super-resolution reconstruction of video frame is used to make processing fast while keeping a high-res video.
- Internet of Things** *Prof. Xiaofan (Fred) Jiang, Columbia University, Sep 2019 - Dec 2019*
- Program ESP8266 to deal with IO, work with APIs, run as server, connect to MongoDB and recognize gesture.
 - Design a distributed scalable system to measure soil conditions over long duration at multiple locations via LoRa wireless communication. Data is stored in MongoDB. A website is built for data visualization and systems control.
- Statistical Learning** *Prof. Predrag Jelenkovic, Columbia University, Sep 2019 - Dec 2019*
- Reproduce the results about identification and authentication in *Smartphone and Smartwatch-Based Biometrics Using Activities of Daily Living* with *WISDM Dataset* using Decision Tree, Random Forest and K-NN.
 - Conduct Support Vector Machine on the same dataset and compare different methods.

TEACHING EXPERIENCES

- Ph.D. Reader for EEC 161 - Applied Probability in Winter 2026
- Ph.D. Teaching Assistant for EEC 256 - Reinforcement Learning in Spring 2025
- Ph.D. Teaching Assistant for ENG 006 - Engineering Problem Solving in Winter 2025
- Ph.D. Teaching Assistant for EEC 173A / ECS 152A - Computer Networks in Winter 2024 and Winter 2023
- M.S. Teaching Assistant for E4764 IoT - Intelligent and Connected Systems in Fall 2020

TECHNICAL SKILLS

- Programming:** Python, C/C++/SystemC, Matlab, HTML, R, VerilogHDL
- Software & Tools:** **Design:** KiCad, Autodesk Inventor, Altium Designer, LabVIEW, FDTD Solutions
- Platform:** VS Code, PlatformIO, etc
- Others:** GitHub, Linux, NS-3, LaTeX, Docker, PyTorch, OpenCV, Spark, AWS, GCP

AWARD

- Columbia University: Nikola Tesla Scholarship *Sep 2019*
- Xi'an Jiaotong Univeristy Third Scholarship *Dec 2018*
- Xi'an Jiaotong Univeristy Siyuan Scholarship *Oct 2017*
- Xi'an Jiaotong Univeristy Outstanding Student *Oct 2017*
- First Prize in China Undergraduate Mathematical Contest in Modeling, Provincial *Dec 2016*
- Second Prize in VEX Robotics China Open, National *Dec 2016*